

However, the quality of the image suffers heavily because of this reduced resolution.

Sensor

The sensor's function is to capture the light coming in through the lens. This information, which is in the form of light, is converted into a digital signal and forwarded to the processor for reconstruction. A high-quality sensor will reproduce a crisp image.

Sensors are of two types: CMOS (Complementary Metal Oxide Semiconductor) and CCD (Charge Coupled Device). Low-cost CMOS sensors are found in low-end cameras. CCDs, on the other hand, are found in most mid-level to prosumer-class cameras. CCDs offer better image quality, but at the same time, are expensive to manufacture. CMOS is cheaper to produce, consumes lesser power, and current improvements in quality have helped it find a place in high-end digital SLR cameras such as the Canon 350D.

The Processing Engine

The camera's processor gets its information from the sensor and starts reconstructing the image. Here, speed is of prime importance, you wouldn't like to sit idle for a full minute while your processor crunches numbers. The start-up time and time interval (cycle time) between two shots depends on the processing engine. Cameras such as the Sony DSC-P100 boot up very quickly, and you don't have to wait long

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after clicking one pic to click the next. The Canon IXUS400, with its DIGIC processor, is another camera with a low cycle time. Cameras with a low cycle time are easy to use when images have to be captured in quick succession, for instance, in wildlife photography.

Memory

The small pieces of plastic that store captured images are called memory cards. The speed at which an image is stored depends on the card's read-write speed. Thus, a card with a fast write speed reduces the time interval required between two shots, and a lower read time means faster image transfer to a PC. MultiMedia Card (MMC), SecureDigital (SD), Compact Flash (CF), xD-Picture card and MemoryStick Pro are some of the widely-used memory card formats. SD cards have one advantage over the others: they are compatible with MMC. Many models, such as the HP Photosmart R707 and the Samsung V5, support both formats.

An important point to note, however, is that the camera writes the data onto the memory card at a certain speed, say 12x. Thus a faster card, with a faster write speed—say 40x—need not mean improved performance if the camera cannot handle it.

The Interface

USB is the interface of choice for digital cameras. The newer generation cameras come with

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